

Stakeholder Needs and Requirements Definition

AI-Assisted Stock Research
and Ranking Platform (StockLens)

Group 29

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1 Introduction

1.1 Purpose of Stakeholder Needs Analysis and Requirements Definition

The purpose of this analysis is to establish whose needs StockLens must address and translate those needs into a traceable basis for system development and evaluation. StockLens supports students and beginning investors in comparing stocks, understanding the factors behind a ranking, and inspecting the evidence used in an explanation. The analysis therefore begins with the people who perform these research tasks, while also considering those responsible for developing, assessing, operating, and supporting the platform.

The resulting stakeholder priorities guide the selection of personas, the refinement of needs, and the definition of stakeholder requirements. Each requirement is connected to an originating need, a measure of effectiveness, and a proposed validation approach. These connections allow later design decisions to be evaluated against an identified stakeholder purpose and help the team recognize features that fall outside the agreed project scope.

1.2 Description of Stakeholder Needs Analysis and Requirements Definition

The analysis builds on the stakeholders, operational concept, and research scenarios established in the Business or Mission Analysis [2]. It follows the Chapter 3 process in the Lab Manual [1], first identifying the roles that use StockLens, influence its delivery, evaluate its outcomes, or control services on which it depends. A Power-Interest Grid then distinguishes these roles according to their ability to affect the project and their interest in its outcomes.

Stakeholders in or near the high-power, high-interest quadrant receive detailed analysis through personas and empathy maps. Their task contexts, concerns, and expected outcomes inform primitive needs. Critical-to-quality considerations clarify what a satisfactory outcome means from each stakeholder's perspective, and effective needs provide the basis for properly formed stakeholder requirements. Measures of effectiveness and validation criteria subsequently define how satisfaction of those requirements will be assessed.

The report preserves stakeholder and need identifiers across these steps so that the reasoning can be followed from stakeholder selection to requirements and supporting model relationships. The personas and PIG placements are scenario-based analytical representations; they do not constitute records of completed stakeholder interviews. Their assumptions should be revisited when user feedback, service choices, or operating responsibilities change.

The statements below define proposed obligations and assessment methods. They do not establish that a prototype meets those obligations. The stored model links and remaining assessment work are reported in Section 4. Acceptance depends on recorded evidence against a confirmed baseline.

2 Stakeholder Needs Definition

The stakeholder baseline is adapted from the team's Business or Mission Analysis [2]. It retains stakeholder identifiers ST-01 to ST-06 and need identifiers N-01 to N-08 so that the two reports can be traced consistently. The personas are fictional analytical scenarios, and the measures and acceptance thresholds are proposed criteria subject to stakeholder review.

2.1 Stakeholders

StockLens involves more than the person viewing a stock comparison. A useful research result depends on understandable output, a feasible implementation, consistent assessment evidence, repeatable operation, and access to external data and AI services. Table 1 identifies six stakeholder groups across these responsibilities.

The groups represent roles rather than necessarily different individuals. A student may contribute to development and later maintain the platform, but the delivery concerns of the project team differ from the operational concerns of a maintainer. Likewise, the market-data, news, and AI services are technical interfaces; the organizations operating those services are the stakeholders considered here.

Table 1: Preliminary stakeholder profiles and the reason for including each group.

Stakeholder / role	Needs and concerns	Why included
ST-01 Students and beginning investors Use and feedback	Consistent comparisons (N-01), understandable ranking drivers (N-02), and inspectable sources and dates (N-03). Concern: unexplained scores, stale evidence and misplaced confidence in AI text.	They perform UC.1 and determine whether the research workflow is useful. Their feedback should guide explanation detail and presentation.
ST-02 Project team Acquisition, development, deployment	A feasible scope within available time, skills and service resources (N-04). Concern: growing integration effort, uncertain access and unconfirmed costs.	The team selects providers, builds the prototype and resolves trade-offs between user benefit and delivery effort.
ST-03 Instructor and evaluators Review and acceptance	Traceable, consistent evidence linking the problem, needs, model and prototype (N-05). Concern: unsupported claims or diagrams that disagree with the implementation.	They assess the course deliverables and need to distinguish a connected demonstration from validated user benefit.
ST-04 System maintainer Support and retirement	Repeatable operation and documented handover or shutdown (N-06). Concern: undocumented configuration, provider changes and abandoned credentials.	Someone must be able to sustain or retire the system. The project team fills this role provisionally; no long-term operator is designated.
ST-05 Market data and news providers External service provision	Use of their services and content within agreed access, retention and display conditions (N-07). Concern: unsupported redistribution or requests outside selected access limits.	The concept depends on their records, timestamps and source identifiers; their conditions constrain the feasible design.
ST-06 External AI service provider External analysis service	Intended inputs through approved service access (N-08). Concern: inappropriate payloads or use outside selected service conditions.	The current boundary places model hosting outside StockLens. This stakeholder must be revisited if that architectural assumption changes.

In the operational concept, ST-01 corresponds to X.01 Research User; ST-04 to X.05 System Maintainer; ST-05 covers X.02 Market Data Provider and X.03 News Provider; and ST-06 corresponds to X.04 AI Model Service. ST-02 and ST-03 are life cycle stakeholders rather than additional participants in the normal research sequence.

Together, these groups cover the user experience, project delivery, independent evaluation, operational continuity, and external dependencies. Their interests can conflict: users may want broader coverage, while the development team must control scope and maintainers must keep operation repeatable. Stakeholder prioritization therefore considers both the value of an outcome and the ability of a role to affect its delivery.

2.2 Personas

The four personas in Figures 1–4 develop the roles established in the Business or Mission Analysis [2]. Their coverage follows the stakeholder register in Table 1 and the PIG-based selection explained in Section 2.3 and Table 4. Together, they represent all four prioritized roles, ST-01 through ST-04, as required by the Lab Manual’s focus on major stakeholders [1].

These are fictional, scenario-based profiles, not records of stakeholder interviews. The representative statements, actions and feelings are analytical hypotheses drawn from the documented research, review and handover situations. Portraits distinguish the profiles; they do not identify research participants. User feedback and operating experience are needed to confirm or revise these hypotheses.

Table 2: Persona coverage of the prioritized stakeholder roles.

Persona	Role	Distinct task perspective	Need link
Maya Chen	ST-01	Explain a two-stock comparison and inspect the evidence before a student discussion.	N-01–N-03
Daniel Brooks	ST-01	Prioritize further research while checking comparability and evidence freshness.	N-01–N-03
Elena Park	ST-03	Follow a requirement from its need to model relationships and evidence status.	N-05
Ben Carter	ST-02; ST-04	Bound the deliverable, then operate and hand it over without undocumented assistance.	N-04; N-06

How the profiles work together. Maya and Daniel represent different situations within ST-01, rather than two additional stakeholder groups. Maya emphasizes explaining the reasoning to others; Daniel emphasizes selecting the next research task from information with a consistent basis. Both require understandable, inspectable results. Elena examines whether the team’s claims can be traced to evidence. Ben examines whether the agreed workflow can be delivered and repeated by another person. His two responsibilities remain separate roles even when one team member performs both.

Reading the empathy maps. Each map connects what the persona says, thinks, does and feels to the same situation described in the profile. Frustrations identify obstacles; the closing interpretation links those obstacles to existing need identifiers. The maps inform needs analysis rather than demonstrate that a proposed feature or acceptance threshold has been validated. ST-05 and ST-06 remain external constraint sources in the PIG; they are not additional priority personas.

Maya Chen — Student researcher



Role and context

Student researcher (ST-01; N-01–N-03).

“Why is Stock A above Stock B? Can I open the article behind that claim?”

A day in the life

Maya is preparing for a student investment discussion. She has two companies in mind and can read their price charts, but she expects someone to ask why one ranks above the other. A polished summary is little help if she cannot explain it in her own words. In the proposed workflow, she starts with the score breakdown, checks the drivers named in the explanation, and opens the article behind a claim before using it in her discussion. For Maya, a useful result is one she can question and explain. This connects a consistent comparison (N-01) with understandable drivers (N-02) and accessible evidence (N-03).

Fears and frustrations

An attractive summary may hide the calculation, an unsupported claim or an old observation.

What would help

A comparison she can reproduce and explain using visible indicators and source dates.

Values and goals

Understand the comparison well enough to explain it to classmates and answer follow-up questions.

Empathy Map

SAYS

Why is Stock A above Stock B?
Can I open the article behind that claim?



THINKS

I need to defend this comparison in a student discussion and understand the reasoning myself.

DOES

Compares two stocks, checks the score components and follows a cited source before preparing her explanation.

Maya Chen

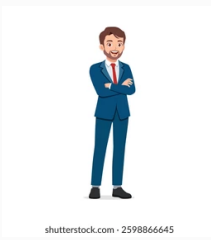
FEELS

Uncertain when the rank is unexplained; more confident when the drivers and evidence are inspectable.

What this means for StockLens. Maya’s uncertainty connects three needs: a consistent comparison (N-01), understandable drivers (N-02), and inspectable evidence (N-03). SR-01–SR-03 carry these into obligations. Her scenario motivates checking comprehension and source inspection, not merely whether text appears on screen.

Figure 1: Maya Chen: a scenario-based persona and empathy map.

Daniel Brooks — Beginning investor



Role and context

Beginning investor (ST-01; N-01–N-03).

“Are both scores based on the same date? Does first place mean this is right for me?”

A day in the life

Daniel opens his research list after work. He wants to decide which company deserves a closer look, and he has little patience for switching between pages just to find out whether two numbers refer to the same date. If one score uses older observations, he wants to see that before drawing a conclusion. The ranking helps him choose what to research next; it does not settle whether a stock suits him. His scenario puts particular weight on a common comparison basis (N-01) and visible source dates (N-03), with N-02 helping him understand what the ranking actually says.

Fears and frustrations

Different data dates can make an apparently simple comparison misleading.

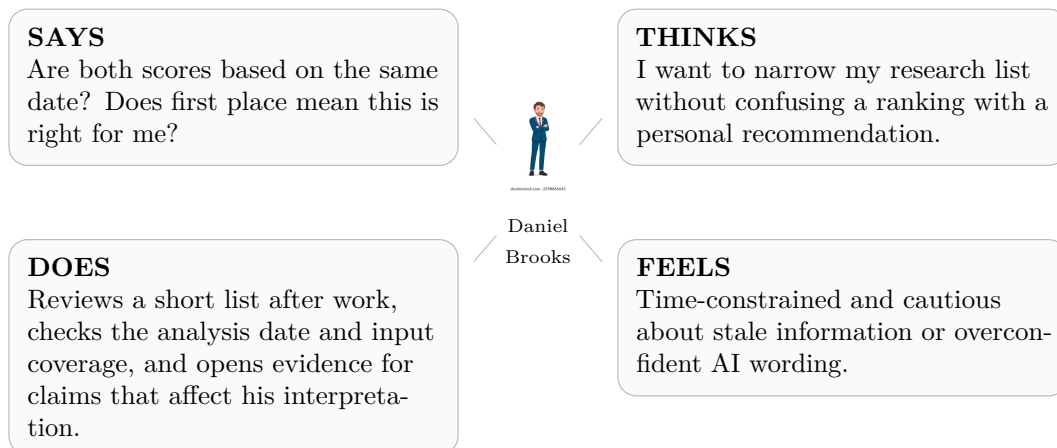
What would help

A consistent research starting point with visible evidence and a clear interpretation of the ranking.

Values and goals

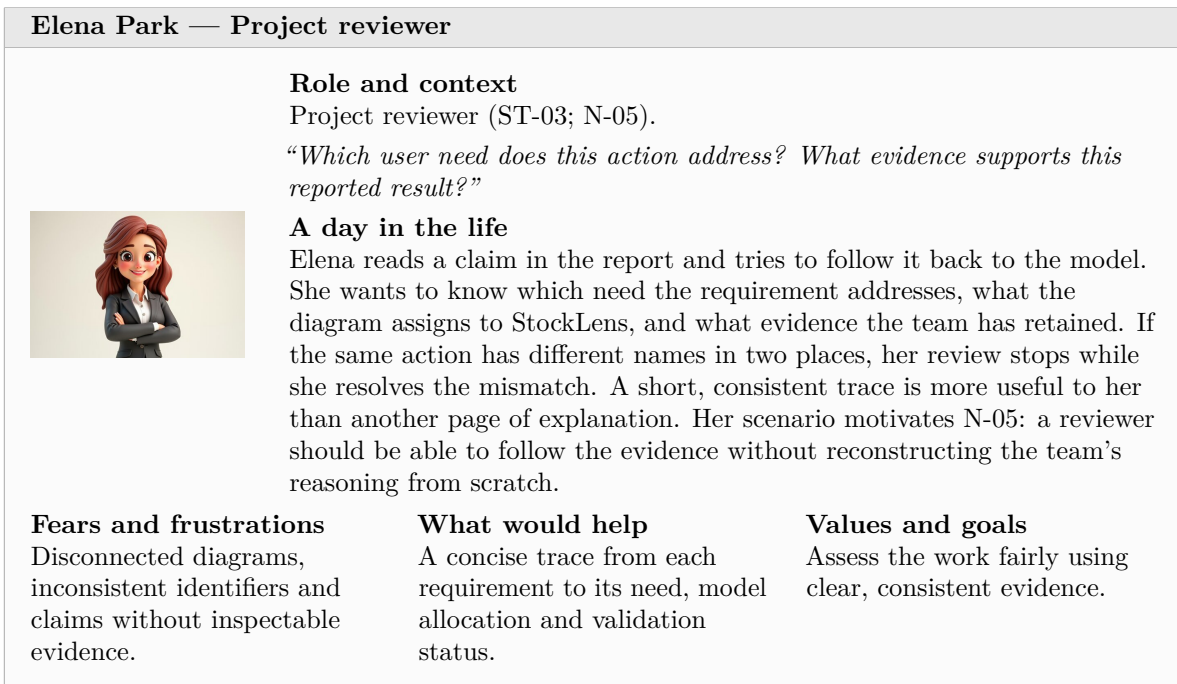
Use limited research time well and stay in control of the next research decision.

Empathy Map

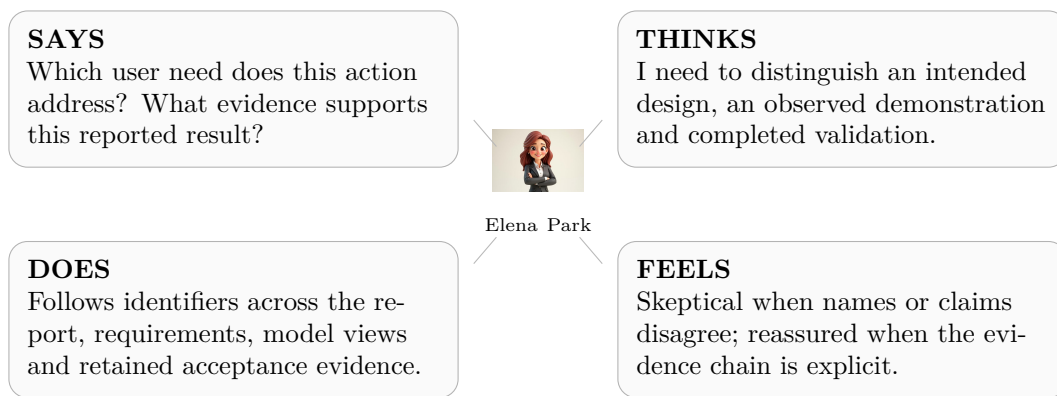


What this means for StockLens. Daniel shares N-01–N-03 with Maya but tests them in a time-limited research setting. His concern about mixed dates supports the common basis in SR-01 and source dates in SR-03. The desire to save time remains a hypothesis; it does not introduce an unconfirmed speed target.

Figure 2: Daniel Brooks: a scenario-based persona and empathy map.



Empathy Map



What this means for StockLens. Elena’s interrupted review exposes the need for a consistent evidence chain (N-05). SR-05 addresses the traceability record, including validation method and evidence status. A planned check must remain distinguishable from a completed check and its retained result.

Figure 3: Elena Park: a scenario-based persona and empathy map.

Ben Carter — Student developer and maintainer



Role and context

Student developer and maintainer (ST-02 and ST-04; N-04, N-06).

“Can we finish the agreed workflow? Could another teammate run it without me?”

A day in the life

Before the demonstration, Ben compares the requested workflow with the team’s available time, skills and service access. He wants an agreed core path that the team can finish and demonstrate (ST-02; N-04). Later, acting as maintainer, he runs a teammate’s version using the handover instructions. A missing configuration step forces him to ask its author for help. He needs the environment and operating steps recorded so that startup, shutdown and transition do not depend on one person (ST-04; N-06).

Fears and frustrations

Scope that exceeds available resources; missing setup steps and unclear ownership at handover.

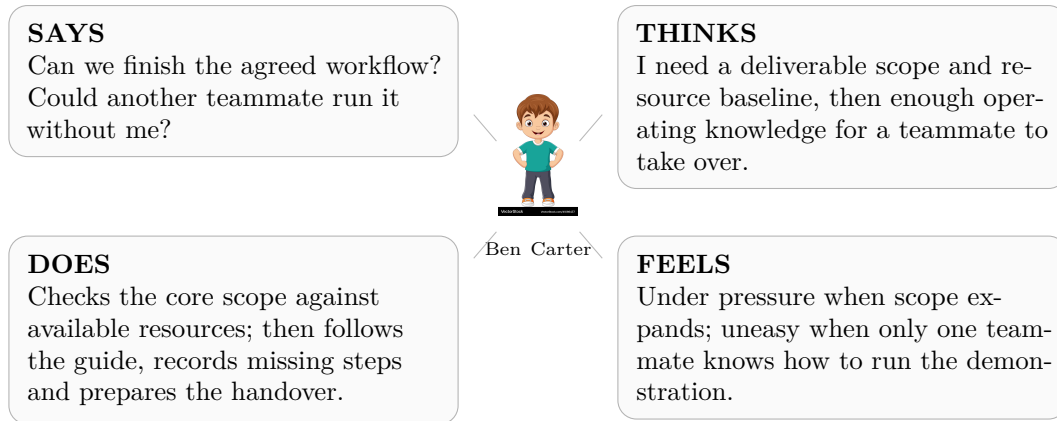
What would help

A feasible core workflow and a repeatable operating procedure that another member can follow.

Values and goals

Keep the demonstration repeatable and make handover manageable for the next teammate.

Empathy Map



What this means for StockLens. The scope concern supports N-04 for ST-02; the missing-step concern supports N-06 for ST-04. SR-04 addresses the bounded demonstration and SR-06 addresses independent operation. Completion of one does not establish the other.

Figure 4: Ben Carter: a scenario-based persona and empathy map.

2.3 Stakeholders Power-Interest Grid

The Power-Interest Grid evaluates stakeholders in the context of the current StockLens course project. **Power** refers to the ability to influence project scope, design, assessment, operation, or access to essential services. **Interest** refers to the extent to which a stakeholder is directly concerned with the outcomes of this particular project. A provider’s importance to the architecture does not automatically imply high interest in the course submission.

Figure 5 and Table 3 use the same six stakeholder identifiers as Table 1. The positions are qualitative rather than measured scores; distances between points do not represent numerical differences. Research users are placed near the boundary of the high-power region: they have limited formal authority over

the project, but their ability to complete and understand the research task is central to assessing operational usefulness.

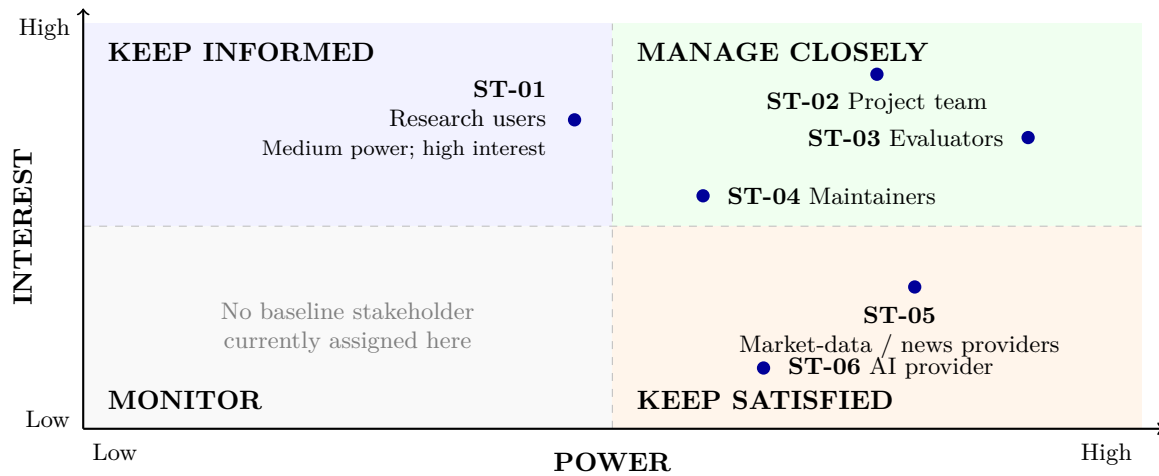


Figure 5: Qualitative Power–Interest Grid for the current project phase. Positions represent analytical judgments, not measured scores.

Table 3: Basis for stakeholder placement and engagement.

Stakeholder	Power	Interest	Rationale and engagement
ST-01 Research users	Medium	High	Directly affected by comparison quality and explanation clarity, but lack formal project authority. Keep informed and actively involve them in task review.
ST-02 Project team	High	High	Controls scope and implementation choices and is responsible for delivery. Manage closely through coordinated decisions.
ST-03 Evaluators	High	High	Apply assignment criteria and assess the coherence and evidence of the submission. Manage closely by aligning review evidence with documented criteria.
ST-04 Maintainers	Medium–high	High	Configuration and operating knowledge affect whether the platform remains usable after initial development. Manage closely through operating and handover reviews.
ST-05 Data/news providers	High	Low	Can restrict access or content use, while having limited direct interest in an individual student project. Keep satisfied through documented, appropriate use.
ST-06 AI-service provider	High	Low	Can constrain explanation delivery through access and service conditions; direct project interest is limited. Keep satisfied through approved use and review of service conditions.

The grid reveals two different sources of influence. The project team, evaluators, and maintainers influence whether StockLens can be delivered, assessed, and operated. They therefore occupy the high-interest region with substantial project power. Research users sit nearby because their tasks define what a useful result must accomplish, even though they do not control the schedule or architecture. Their lower formal authority is a reason to provide deliberate opportunities for feedback rather than to reduce attention to their needs.

External providers occupy the high-power, lower-interest region. They can constrain access to the data or explanation services required by the workflow, but the success of this specific course project is unlikely to be a major concern for them. Their influence is addressed through dependency management and review of applicable service conditions. Consequently, the grid supports intensive needs analysis for

the human roles closest to project outcomes, alongside focused management of external constraints. The lower-power, lower-interest quadrant remains empty because none of the six baseline roles is currently assigned there; additional stakeholders should be added when supported by the project context.

2.4 Stakeholder Prioritization

Following the Lab Manual’s focus on stakeholders in or near the upper-right quadrant [1], detailed persona and needs analysis concentrates on ST-01 through ST-04. These roles jointly define whether StockLens is useful to its intended users, feasible for the team to deliver, assessable against course expectations, and maintainable after the initial demonstration.

ST-01 is included despite having less formal power than the project team or evaluators. Excluding the intended users would leave the analysis without a direct basis for evaluating comparison quality, explanation comprehension, and evidence inspection. ST-02 and ST-04 remain distinct stakeholder roles even when performed by the same person, because successful development and successful handover require different information.

Table 4: Priority stakeholders and connections to subsequent analysis.

Selected role	Reason for inclusion	Persona and need connection
ST-01 Research users	Their research tasks define the intended operational value.	Maya and Daniel; N-01–N-03: comparison, comprehension, and evidence inspection.
ST-02 Project team	Controls delivery decisions and must keep the project feasible.	Ben’s development perspective; N-04: manageable delivery.
ST-03 Evaluators	Determine whether the submission meets stated assessment expectations.	Elena; N-05: traceable reasoning and evidence.
ST-04 Maintainers	Determine whether operation and handover can be repeated by another person.	Ben’s maintenance perspective; N-06: operational continuity.

ST-05 and ST-06 remain in the stakeholder register and PIG because their service conditions constrain the selected workflow. They are not selected for detailed personas in this iteration. N-07 and N-08 are retained as supporting external-service needs and constraints, rather than evidence of high provider interest. They support the prioritized team and maintainer roles (ST-02 and ST-04), whose feasible delivery and repeatable operation depend on controlled service use. The provider perspective identifies the source of a constraint; it does not extend the detailed-persona scope beyond the selected roles. Their interpretation depends on the documented conditions of the services selected for the project. Changes in provider access, deployment arrangements, or stakeholder responsibilities should trigger a review of the grid and the resulting priorities.

2.5 Stakeholder Needs

This section refines the existing N-01–N-08 baseline from the Business or Mission Analysis [2]. It uses the PIG priorities in Table 4, the persona coverage in Table 2, and the documented comparison, review and handover situations. This follows the Lab Manual’s progression from primitive concerns through critical-to-quality considerations to stakeholder need statements [1]. It is an analysis of the existing project material, not a claim that new interviews or surveys have been completed.

N-01–N-03 address the effectiveness of the research task for ST-01. N-04–N-06 address delivery, assessment and continued operation for the other prioritized roles, ST-02–ST-04. N-07 and N-08 preserve the external-service constraints identified in the earlier baseline: their source perspectives are ST-05 and ST-06, while their practical consequences affect the project team and maintainer. They do not add providers to the priority-persona set.

2.5.1 Primitive Needs

Primitive needs retain the stakeholder’s underlying concern before specifying system behavior. The first-person statements in Table 5 are analytical paraphrases, not interview quotations. The source column identifies the situation that motivates each concern; it does not constitute evidence that the need has been validated.

Table 5: Primitive needs and their scenario-based origins.

ID / perspective	Primitive need	Context and origin
PN-01 / ST-01	I need a fair way to compare the stocks I am researching.	Maya’s two-stock discussion and Daniel’s research list both depend on a comparable basis.
PN-02 / ST-01	I need to understand why one stock ranks above another.	Maya must explain the result; Daniel needs to interpret what the ranking means.
PN-03 / ST-01	I need evidence I can check before relying on a claim.	Maya follows a cited article; Daniel checks whether the supporting information is current enough for his research.
PN-04 / ST-02	We need a project we can deliver with the resources we have.	Ben’s development perspective: an expanding scope can exceed available time, skills or service access.
PN-05 / ST-03	I need to follow the reasoning and evidence behind the submission.	Elena’s review is interrupted by inconsistent names, missing links or unsupported claims.
PN-06 / ST-04	I need to keep the demonstration usable without relying on its original author.	Ben’s maintenance perspective: missing setup knowledge prevents independent operation and handover.
PN-07 / ST-05	We need our services and content used within agreed conditions.	Baseline provider-use concern; ST-02 and ST-04 depend on access and permitted use. No provider interview is claimed.
PN-08 / ST-06	We need appropriate inputs for the requested analysis.	Baseline AI-service input concern; ST-02 and ST-04 manage the request boundary. No provider agreement is inferred.

2.5.2 Critical-to-Quality Considerations

A critical-to-quality consideration describes what a satisfactory outcome means to the stakeholder. Table 6 makes these qualities explicit without turning the needs into a design specification. Each CTQ identifier has the same numeric suffix as its primitive and effective need, preserving the refinement path PN-01 → CTQ-01 → N-01, and so on.

Table 6: Quality considerations used to refine the primitive needs.

CTQ / need	Quality that matters	What a satisfactory outcome means
CTQ-01 N-01	Comparability	Maya and Daniel can interpret both stocks against the same analysis date and documented indicator basis. Different observation dates remain distinguishable rather than being presented as equivalent.
CTQ-02 N-02	Understandable, grounded reasoning	The user can relate the explanation to the drivers and values behind the ranking. Fluent wording alone is insufficient if it contradicts the calculation or leaves the reason unclear.
CTQ-03 N-03	Evidence traceability and freshness	The user can inspect the source supporting a news claim and distinguish its publication date from the comparison’s analysis and market-data dates. The source must support the claim, not merely be linked.
CTQ-04 N-04	Delivery feasibility	Ben and the team can demonstrate an agreed core research workflow within a confirmed time and resource envelope. More features do not improve suitability if the core path cannot be delivered.
CTQ-05 N-05	Reviewable traceability	Elena can follow a consistent chain from a requirement to its originating need, relevant model entities and evidence status. Planned validation remains distinguishable from a completed result.
CTQ-06 N-06	Operational continuity	A team member other than the original author can follow the documented operating process. Handover identifies the knowledge and responsibilities needed for continued use or orderly shutdown.
CTQ-07 N-07	Controlled service and content use	The team can relate planned access, retention and display to the selected providers’ documented conditions. Public availability alone does not establish permission for the intended use.
CTQ-08 N-08	Appropriate analysis inputs	The analysis request contains the agreed task-relevant inputs. The approved input schema distinguishes that content from authentication and excludes credentials or unnecessary personal information.

These qualities guide the measures and methods in Sections 3.2 and 3.3. They do not establish new numerical acceptance targets. The research universe, scoring basis, resource envelope, provider conditions and AI input schema must be confirmed where the later requirements depend on them. The inherited user-task thresholds remain proposed criteria, not findings from the personas.

2.5.3 Effective Needs

The statements in Table 7 refine the primitive concerns into specific stakeholder outcomes while preserving N-01–N-08. They remain in stakeholder language; the corresponding *shall* obligations are defined separately in Table 8. Each row carries one principal outcome and identifies its downstream requirement, making the connection visible in the report as well as available for model traceability.

Table 7: Effective stakeholder needs and their refinement and requirement links.

Need / perspective	Effective need statement	Refinement / trace
N-01 / ST-01	I need to compare two stocks on a common analysis date using a consistent indicator basis.	PN-01; CTQ-01 → SR-01
N-02 / ST-01	I need to understand a displayed ranking through the computed drivers that explain the difference between the stocks.	PN-02; CTQ-02 → SR-02
N-03 / ST-01	I need inspectable, dated evidence for the news claims used in my stock comparison.	PN-03; CTQ-03 → SR-03
N-04 / ST-02	We need an agreed core research workflow that our team can deliver within its available time, skills and service resources.	PN-04; CTQ-04 → SR-04
N-05 / ST-03	I need a traceable account of how each stakeholder requirement follows from a need and is supported by model and assessment evidence.	PN-05; CTQ-05 → SR-05
N-06 / ST-04	I need continuity of operation through instructions that another team member can follow without the original author's undocumented knowledge.	PN-06; CTQ-06 → SR-06
N-07 / ST-05	We need the project's use of our services and content to remain within the conditions applicable to the selected access arrangements.	PN-07; CTQ-07 → SR-07
N-08 / ST-06	We need the analysis request to contain only the inputs agreed for that task through the approved service access.	PN-08; CTQ-08 → SR-08

Connection to the prioritized roles. N-01–N-03 express the concerns shared by Maya and Daniel; N-04 captures Ben's delivery perspective, N-05 Elena's review perspective, and N-06 Ben's operating perspective. N-07 and N-08 retain their provider source perspectives for continuity with the earlier report, while supporting feasible delivery and operation for ST-02 and ST-04. They are supporting constraints rather than evidence of high provider interest in the course project.

Limits of the current refinement. These links record intended coverage, not proof that a need has been satisfied. Section 3.1 carries operational continuity into SR-06's documented transition procedure and controlled service use into SR-07's conformance obligation. Their assessment must cover those outcomes: a startup check or a completed conditions record alone is insufficient. Provider conditions and the request schema require confirmation against the selected services.

3 Stakeholder Requirements Definition

3.1 Stakeholder Requirements

The eight requirements in Table 8 translate the effective needs in Section 2.5.3 into proposed obligations. The refinement follows the Lab Manual's requirement-statement and traceability activities [1]. Each statement names its responsible subject, uses *shall*, and retains its originating need. A trace link establishes the reason for an obligation; it does not establish that the obligation has been met.

SR-01–SR-03 and SR-08 specify StockLens behavior. SR-04–SR-07 specify delivery, evidence and support obligations of the StockLens project. This distinction preserves the lifecycle concerns of ST-02–ST-04 without assigning project-management duties to the software or an external provider. N-07 and N-08 remain supporting constraints on the prioritized workflow, consistent with Section 2.3.

Table 8: Stakeholder requirements, originating needs and rationale.

ID / need	Requirement statement	Rationale
SR-01 / N-01	StockLens shall display a comparison of two eligible stocks using a common analysis date and the same documented indicator and scoring configuration.	Maya and Daniel need a common basis for interpreting the two results.
SR-02 / N-02	StockLens shall provide an explanation of each displayed ranking that identifies the ranking’s main computed drivers and their corresponding indicator values.	The explanation must connect the user’s interpretation to the calculation, rather than substitute AI wording for scoring.
SR-03 / N-03	StockLens shall associate each displayed news claim with an inspectable supporting source identifier and publication date, alongside the analysis and market-data dates for the comparison.	Inspectable evidence and distinct dates help the user assess support and freshness.
SR-04 / N-04	The StockLens project shall deliver an executable instance of the agreed UC.1 research workflow within the time and service-resource limits recorded in the demonstration baseline.	Ben’s delivery concern requires a bounded workflow that the team can finish with its available resources.
SR-05 / N-05	The StockLens project shall maintain a traceability record for each stakeholder requirement identifying its originating need, related model entities, validation method and evidence status.	Elena needs to follow the reasoning and distinguish planned assessment from completed evidence.
SR-06 / N-06	The StockLens project shall provide an operating and transition guide enabling a team member other than its author to start, run and shut down the demonstration and carry out the documented handover or retirement procedure without undocumented instructions.	Ben’s maintenance concern extends beyond startup to continuity, transfer of responsibility and orderly shutdown.
SR-07 / N-07	The StockLens project shall conduct its use of each selected market-data and news service in accordance with the applicable access, retention and display conditions identified in its dated service-use record.	Recording conditions alone does not meet the need; actual service and content use must conform to them.
SR-08 / N-08	StockLens shall restrict AI analysis-request content to the approved field schema, excluding credentials and unnecessary personal information from that content.	Analysis inputs are limited to the explanation task; service authentication remains separate from analysis content.

Meaning and scope of the statements. The inherited BMA scope is a research-only comparison service covering up to 100 predefined U.S. stocks with daily market observations; automated trading, order execution, personalized portfolio management and intraday support remain outside this baseline [2]. The exact eligible list and scoring configuration still require versioned confirmation. An *eligible stock* in SR-01 belongs to that agreed research universe and has the inputs required by the scoring configuration. A common analysis date identifies the comparison; it does not imply that all source observations were published at the same time. The market-data date identifies the observations used, while the publication date identifies the news item. Missing or differently dated inputs cannot be treated as an equivalent basis without an explicit rule in that configuration.

For SR-02, *main computed drivers* means the factors selected from the actual score breakdown under the documented driver-selection rule. The rule and the scoring configuration must be fixed before the comprehension task is assessed. The explanation describes those calculations; it does not assert future returns. For SR-03, a source identifier must allow inspection of the supporting item, and that item must support the associated claim. A citation’s presence alone is insufficient.

The SR-06 guide covers the documented environment, dependencies, operating steps and the selected transition path. Its handover or retirement procedure identifies responsibility, configuration, retained

records and treatment of service access. This makes the existing transition-checklist concern explicit without prescribing a new implementation. For SR-07, the dated service-use record identifies the source of each applicable condition and the corresponding use decision. A review of planned use is preparation for acceptance, not evidence that all subsequent operation conforms.

Visible need-to-requirement traceability. Table 9 preserves the stakeholder and scenario links established in Sections 2.2 and 2.5. The same IDs are retained in the Innoslate Needs and Stakeholder Requirements documents. Each requirement has a stored traced-from link to its originating need. This records the source of the obligation; it does not establish that the need has been satisfied.

Table 9: Origin and intended coverage of the requirement set.

Requirement	Origin	Stakeholder concern carried forward
SR-01	N-01 / ST-01	Maya and Daniel: comparable two-stock research results.
SR-02	N-02 / ST-01	Maya and Daniel: an understandable reason for the ranking.
SR-03	N-03 / ST-01	Maya and Daniel: inspectable, dated support for news claims.
SR-04	N-04 / ST-02	Ben as developer: feasible delivery of the agreed core workflow.
SR-05	N-05 / ST-03	Elena: a reviewable chain of reasoning and evidence status.
SR-06	N-06 / ST-04	Ben as maintainer: independent operation and documented transition.
SR-07	N-07 / ST-05	Provider conditions constrain service use by ST-02 and ST-04.
SR-08	N-08 / ST-06	The agreed analysis-input boundary constrains requests made by StockLens.

Baseline decisions and acceptance boundary. The set retains eight identifiers and one principal obligation per statement. Before acceptance, the team must confirm the research universe, indicator/scoring configuration and driver-selection rule (SR-01–SR-02); UC.1 scope, environment and time/resource envelope (SR-04); transition path and responsibilities (SR-06); provider conditions (SR-07); and versioned AI input schema and service-access arrangement (SR-08). These are unresolved baseline decisions, not invented numerical targets or completed approvals.

Sections 3.2 and 3.3 provide the proposed measures and evidence methods. In particular, SR-06 requires evidence for the selected transition procedure as well as startup and shutdown; SR-07 requires checking actual use against the recorded conditions, not merely counting completed records. Those coverage checks must be reflected in the final assessment. Each assessment should identify the requirement version, configuration, reviewer, date and result so that later changes remain reviewable.

3.2 Measures of Effectiveness

The measures in Table 10 evaluate the outcomes behind SR-01–SR-08, whose full statements and rationale appear in Table 8. They follow the Lab Manual’s validation and traceability activities [1]. Each requirement has one principal measure. SR-01–SR-03 address research-task effectiveness; SR-04–SR-08 address the delivery, review, continuity and service-use conditions that make the workflow suitable for the prioritized stakeholders.

For Maya and Daniel, a displayed comparison is useful only if they can interpret it, explain the ranking and inspect its evidence. These are separate outcomes: success at one task cannot compensate for failure at another. For Ben and Elena, the corresponding questions are whether the workflow can be delivered, reviewed and operated independently. Provider-related measures retain the constraint sources ST-05 and ST-06 while supporting ST-02 and ST-04.

Table 10: Principal measures and reasons for their selection.

ID / requirement	Measure and unit	Why this measure fits the need
MOE-01 SR-01	Comparison success: participants completing the comparison correctly and unassisted on their first scored attempt divided by participants attempting that task, as a percentage.	N-01 concerns a usable comparison, not merely the presence of two stock records.
MOE-02 SR-02	Driver comprehension: participants identifying both requested computed drivers correctly and unassisted on their first scored attempt divided by participants attempting that task, as a percentage.	N-02 requires understanding of the ranking. Explanation fluency is not a substitute.
MOE-03 SR-03	Evidence-location success: participants retrieving the source and identifying all three dates correctly and unassisted on their first scored attempt divided by participants attempting that task, as a percentage.	N-03 requires inspectable evidence and an ability to distinguish its dates.
MOE-04 SR-04	Core demonstration completion within the confirmed time and service-resource envelope: pass/fail for the agreed UC.1 baseline.	N-04 is about feasible delivery. Extra features cannot offset an incomplete core workflow.
MOE-05 SR-05	Traceability coverage: requirements with complete, consistent need, model, method and evidence-status records divided by all baselined requirements, as a percentage.	N-05 requires a reviewable evidence chain; a link to the wrong entity does not count.
MOE-06 SR-06	Independent operating and transition completion: pass/fail for all documented stages on the selected handover or retirement path.	N-06 requires continuity beyond startup, without reliance on the author's unstated knowledge.
MOE-07 SR-07	Service-use review coverage: providers with documented conditions and completed checks of actual access, retention and display divided by all selected providers, as a percentage.	N-07 concerns use within conditions. Coverage is meaningful only with a conformance check.
MOE-08 SR-08	Payload conformance: conforming analysis requests divided by all requests in the recorded assessment set, as a percentage.	N-08 concerns the request boundary; the inputs must match the agreed schema.

Counting convention. For each user task, let n be the number of participants who begin that task and s the number succeeding without assistance on their first scored attempt. Report s/n and $100s/n\%$. Count each participant once per task; an incorrect, abandoned or assisted first attempt remains in n but not s . Record subsequent practice or retries separately. This proposed protocol preserves the BMA's participant-level denominator. A zero denominator is *not assessed*. Conformance inspections remain separate from user-task success.

3.3 Validation Criteria

Tables 11 and 12 form the stakeholder assessment summary: each row joins the requirement, originating need, measure, proposed acceptance condition and evidence method. The 80% user-task targets are inherited planning targets from the earlier operational concept [2]; they are not measured results or externally established benchmarks. The team must confirm them, the participant sample and the scoring rubric before assessment. No validation is claimed to have been performed.

Table 11: Research-user requirements: measures, criteria and evidence.

Requirement / need / perspective	Measure and proposed acceptance condition	Method and retained evidence
SR-01 / N-01 ST-01 Consistent two-stock comparison.	MOE-01: $\geq 80\%$ unassisted correct completion of the agreed comparison task. Conformance check: both stocks are eligible and use the same analysis date and documented indicator/scoring configuration.	User task + inspection. Ask each participant to compare two stocks using two agreed indicators and the displayed ranking. Score against a prepared answer rubric. Inspect the dated inputs and configuration. Retain answers, assistance records, stock eligibility, dates and configuration version.
SR-02 / N-02 ST-01 Understand computed ranking drivers.	MOE-02: $\geq 80\%$ correctly identify both requested drivers without assistance. Conformance check: every explanation in the assessment set identifies the selected computed drivers and their correct values, with no contradiction of the score breakdown.	User task + calculation review. Ask why Stock A ranks above Stock B. Use a case with two identifiable drivers selected by the frozen rule. Compare answers and explanation content with the checked factor values, weights and score breakdown. Retain the rubric, outputs, input values and discrepancy log.
SR-03 / N-03 ST-01 Inspect source support and dates.	MOE-03: $\geq 80\%$ locate the requested supporting item and distinguish its publication date from the comparison's analysis and market-data dates without assistance. Conformance check: every news claim in the assessment set has an inspectable source that supports the claim and the required date information.	User task + source review. Ask participants to open a displayed claim's source and identify all three dates. Independently inspect each claim and its cited material. Retain answers, source identifiers, dates, the supporting passage or reference, and any inaccessible or unsupported items.

Interpretation. A user-task target and its conformance check are both necessary for a proposed pass. An understandable explanation can still be incorrect; an accurate explanation can still be incomprehensible to the intended user. Finding a citation does not establish that it supports the claim. Report these results separately rather than combining them into one average score.

Identify the assessment set before review. Universal checks in this table refer to that recorded set; a finite pilot does not prove correctness for every possible stock or future output. Broader conclusions require broader evidence. Any exploratory comparison with separate source pages remains outside the acceptance decision until its method and target are agreed.

Table 12: Supporting requirements: measures, criteria and evidence.

Requirement / need / perspective	Measure and proposed acceptance condition	Method and retained evidence
SR-04 / N-04 ST-02 Feasible core delivery.	MOE-04: pass only when the agreed UC.1 workflow executes in the documented environment within confirmed time and service-resource limits. Undefined limits make it not assessable, rather than satisfied.	Demonstration + review. Freeze scope and limits before the run. Record the outcome, environment/version, elapsed time and relevant service-resource use against that baseline. Retain the run record and deviations.
SR-05 / N-05 ST-03 Reviewable traceability.	MOE-05: 100% of the current eight requirements have correct need links, related model entities, a validation method and an explicit evidence status. Recalculate the denominator if the baseline changes.	Inspection. Follow each link in the register, report and Innoslate model; check entity IDs and consistency. Retain reviewed records and discrepancies. A planned status is valid; an invented or broken model link is not.
SR-06 / N-06 ST-04 Independent operation and transition.	MOE-06: pass only when a member other than the guide's author completes startup, the demonstration, shutdown and the selected documented handover or retirement procedure without undocumented instructions.	Observed demonstration + review. Use a controlled project environment. Record stages, assistance and missing instructions. Check responsibility, configuration, retained records and applicable access-transfer or revocation steps. Retain the guide version and executed transition record; a checklist alone is insufficient.
SR-07 / N-07 ST-05; supports ST-02/ST-04 Controlled service use.	MOE-07: 100% provider coverage, with no unresolved discrepancy between applicable conditions and observed access, retention or display in the assessment scope. A conditions register alone does not pass.	Conditions and actual-use inspection. Compare dated provider references with request records, retained data and displayed content. Retain the condition-to-use checklist and decisions for each provider. Missing conditions or unobserved use remain unresolved; planned use is not actual-use evidence.
SR-08 / N-08 ST-06; supports ST-02/ST-04 Intended analysis inputs.	MOE-08: 100% of requests in the recorded set conform to the approved schema, with no credentials or unnecessary personal information in analysis content. Approved service access is a separate prerequisite for assessing N-08; content conformance alone does not close its access-coverage item.	Payload inspection. Compare captured requests with the versioned schema and task purpose. Record the assessment set and deviations. Retain redacted examples and access/configuration evidence without exposing secrets. Review authentication separately from analysis content.

The coverage targets operationalize the universal obligations in the requirements; they are not empirical performance claims. No selected service, no inspected requests or no demonstrated transition means *not assessed*, not 100% success. A scoped review does not establish continuing conformance after conditions, configuration or behavior change.

3.3.1 Proposed Assessment Procedure

1. Confirm the baseline. Resolve the decisions in Section 3.1: research universe, scoring and driver-selection rules, UC.1 scope, environment, time/resource envelope, transition path, provider conditions and AI schema. Record versions and the team decision. Confirm user-task targets before collecting results; do not change a target after seeing the outcome merely to obtain a pass.

2. Prepare participants and cases. Recruit from the student and beginning-researcher group represented by Maya and Daniel. Agree and record sample size and relevant experience. Freeze task instructions, assistance rules and the answer rubric. Use the same checked two-stock case, dated dataset and scoring configuration, with identifiable drivers and inspectable news evidence. Identify outputs and requests to be inspected. The fictional personas describe the intended group; they are not participants or collected evidence.

3. Observe research tasks. Administer each task without coaching and score each participant's first attempt. Record answers and assistance; assisted, abandoned or incorrect first attempts remain unsuccessful in the participant denominator. Log retries separately. Record only exclusions permitted by the pre-agreed protocol, with reasons; do not relabel task failure as exclusion. Report s/n and $100s/n$ percent per task. If $n = 0$, report not assessed.

4. Inspect supporting evidence. Check calculations, sources and dates against displayed outputs. Demonstrate the bounded workflow for SR-04, follow all eight trace records and the inherited skeleton-participant mapping for SR-05, and observe a second member completing the SR-06 operating and transition path. For SR-07, compare actual requests, retained records and displays with conditions. For SR-08, inspect the recorded payload set against the schema. Retain discrepancies after correction and repeat affected checks against the revised version.

5. Decide separately for each requirement. Use *pass*, *fail* or *not assessed*, supported by the applicable measure and conformance checks. Unconfirmed criteria or missing evidence prevent a pass. Report counts, configuration, participant characteristics and limitations. Do not average the eight measures into a single score that could conceal a failure. A small pilot supports conclusions about its recorded cases and participants, not all future use.

Table 13: Fields to retain in each assessment record.

Record field	Content to retain
Identity and baseline	SR, N and MOE IDs; requirement and criterion versions; configuration or document versions.
Assessment scope	Date, reviewer, participant or operator code; cases, providers or requests included; exclusions and reasons.
Observed result	Numerator/denominator or pass/fail observations; assistance; clause-level inspection results and discrepancies.
Evidence and decision	Evidence references; pass/fail/not assessed with rationale; unresolved items; correction and reassessment references.

Current evidence status. This section defines an assessment plan. It does not report recruitment, completed demonstrations, measured success rates or validated system outcomes. Stored model relationships are described in Section 4. Final acceptance depends on the confirmed baseline and retained results. Changes to a requirement or its operating conditions require review of the corresponding need, measure and method.

4 Stakeholder Requirements Management

The Innoslate project *AI-Assisted Stock Platform* maintains the eight effective needs as Statements in SL-N (entity 216537) and the eight stakeholder requirements in SL-SR (entity 216546). Requirement text and rationale match Section 3.1. Each SR has a saved *traced from* link to its corresponding N and a *sourced by* link to SL-SR. The 1 October 2026 revision preserves those identifiers while correcting the intended satisfaction allocations.

SR-02 is allocated to UC.1.6, *Synthesize Grounded AI Explanation*, and UC.1.7, *Render Comparison Dashboard*; SR-03 is allocated to UC.1.7. UC.1.8 remains the user's inspection step, but is no longer used as the satisfying action for system provision of explanations or evidence. SR-04 retains UC.1

and adds A-04, *Deliver Bounded Research Demonstration*, to assign responsibility for delivery within recorded limits. A-05–A-08 retain their requirement links and now have explicit performing Assets. These are design decisions recorded in the model, not claims of implemented or validated performance.

4.1 Requirements Diagram

Continuity with the BMA baseline. The Business or Mission Analysis already contains the Use Case, Action, Activity, Sequence and behavioral Spider views [2]. The current model retains their parent entities and action identifiers. The formal Use Case view is rooted at entity 199938; the Action, Activity, Sequence and behavioral Spider views use UC.1 (entity 185145). They describe the same research workflow rather than separate implementations.

BMA view	Baseline figure	Relationship to this revision
Use Case	Figure 7	Preserve the research user and comparison use case; lifecycle delivery and review roles are not new nominal user interactions.
Action / Activity	Figures 8–9	Preserve the nine-action flow, including the two provider returns. UC.1.7 now explicitly describes driver values, claim–source associations and dates required by SR-02–SR-03.
Sequence	Figure 10	Preserve the participant exchanges. UC.1.8 remains terminal human inspection and does not require a fabricated outgoing message. Before an external AI request, UC.1.6 depends on A-08; this is an explicit model precondition, not evidence of executable gating.
Behavioral Spider	Figure 11	Retain the UC.1 decomposition and performer context. The requirement-centered Spider below serves a different purpose: tracing a particular obligation to its origin, source and allocated behavior.

The request-control dependency does not by itself add a control-flow gate to the Activity view or an additional message to the Sequence view. Those nominal views remain the BMA baseline. The next model revision must show the request check before transmission, its pass/fail outcome, and the behavior when content is rejected, then reconcile the Action, Activity and Sequence views. These are outstanding modeling and implementation tasks; the dependency alone does not complete them. Lifecycle Actions A-04–A-07 are kept separate from the user’s nominal comparison sequence.

Table 14 gives the complete need–requirement–behavior overview before the individual exports. Stakeholder IDs are report-level source associations, not evidence of an additional stored relationship. The dates and statuses follow the review record described at the start of this report.

Table 14: Need, requirement and intended behavior coverage.

Req.	Need	Stakeholder	Satisfied by	Allocation status
SR-01	N-01	ST-01	UC.1	Stored; comparison evidence pending.
SR-02	N-02	ST-01	UC.1.6; UC.1.7	Stored; explanation assessment pending.
SR-03	N-03	ST-01	UC.1.7	Stored; source and date review pending.
SR-04	N-04	ST-02	UC.1; A-04	Stored; delivery and resource evidence pending.
SR-05	N-05	ST-03	A-05	Stored; traceability-record inspection pending.

Req.	Need	Stakeholder	Satisfied by	Allocation status
SR-06	N-06	ST-04	A-06	Stored; independent operation and transition demonstration pending.
SR-07	N-07	ST-05	A-07	Stored; actual-use assessment pending.
SR-08	N-08	ST-06	A-08	Stored; payload inspection pending.

Complete need–requirement coverage. Figures 6–13 are native Innoslate Requirement Diagram, one per requirement group. Together they show all eight originating needs, requirement statements and intended satisfying Actions. The dashed *trace* connection identifies the need; the *satisfy* connection assigns behavior. Draft status remains visible. The figures use the original PNG exports with only blank outer margins clipped in the page layout. Some native connector labels overlap nodes; Table 14 and the accompanying descriptions make their intended endpoints explicit. Correcting the native routing requires a model re-export. The measurable criteria appear in Section 3.3.

SR-01: Comparable stock results. SR-01 traces to N-01 and is allocated to UC.1, Compare Stocks & Evaluate Ranking. This preserves Maya and Daniel’s need for a common comparison basis. MOE-01 still requires a controlled comparison case and user-task evidence.

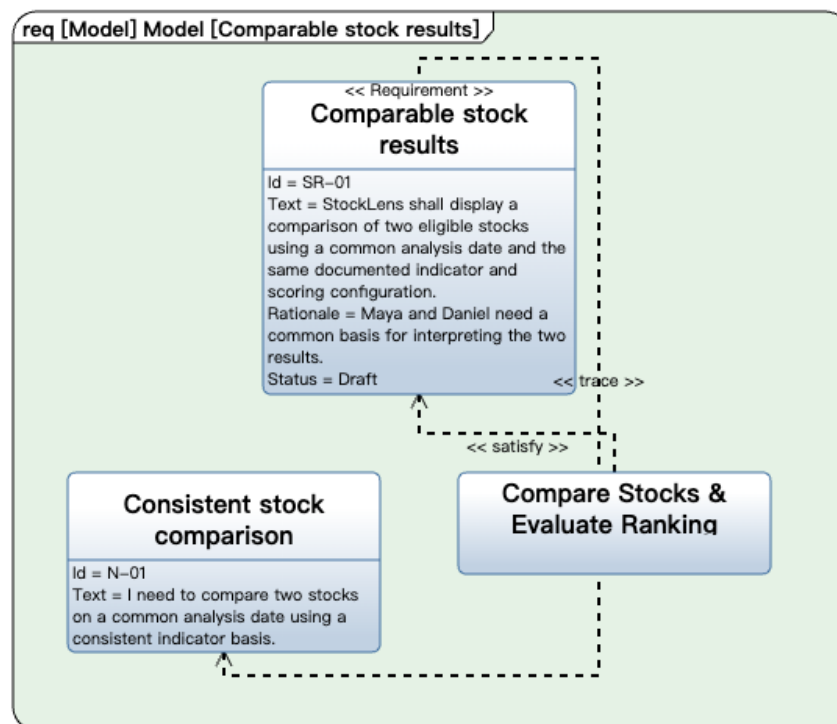


Figure 6: Native Innoslate Requirement Diagram for SR-01.

SR-02: Ranking explanation. SR-02 traces to N-02. UC.1.6 supplies the grounded explanation and UC.1.7 presents it with the computed drivers and their indicator values. The previous allocation to human inspection (UC.1.8) was removed: inspecting an explanation is distinct from providing it. MOE-02 evaluates the explanation against the calculation and the user task.

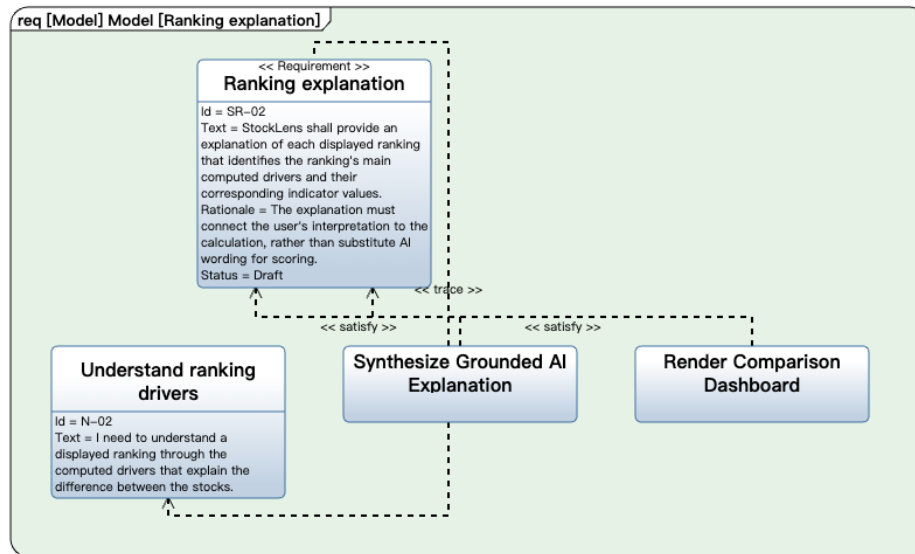


Figure 7: Native Innoslate Requirement Diagram for SR-02.

SR-03: Dated supporting evidence. SR-03 traces to N-03 and is allocated to UC.1.7, Render Comparison Dashboard. Its saved description now requires claim–source associations, publication dates, the analysis date and market-data dates. MOE-03 checks those associations and whether users can inspect the evidence.

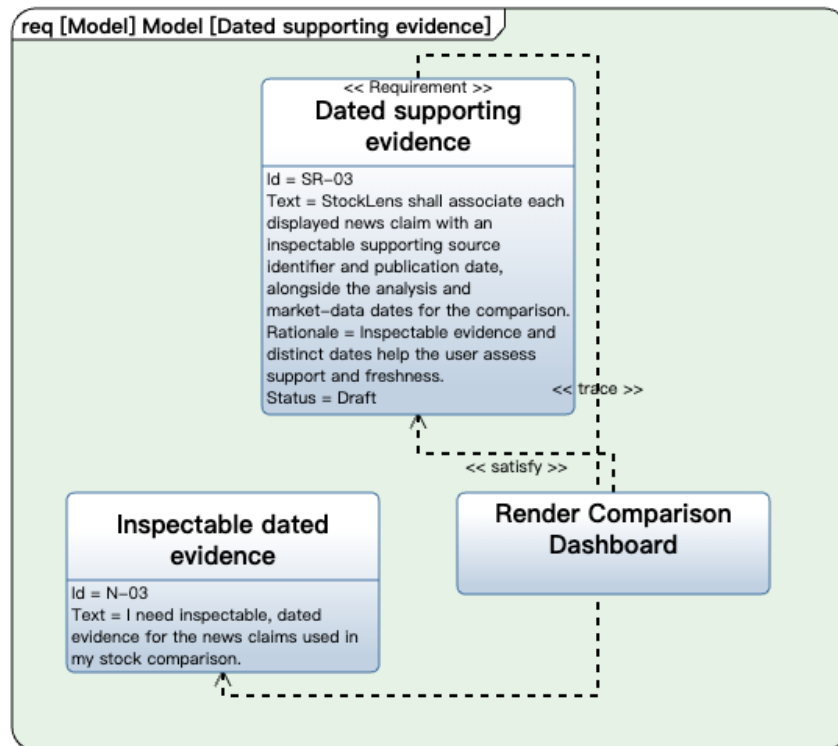


Figure 8: Native Innoslate Requirement Diagram for SR-03.

SR-04: Executable research workflow. SR-04 traces to N-04. UC.1 specifies the research behavior; A-04 assigns the project team responsibility for delivering an executable instance within the versioned demonstration limits. Both allocations are shown because workflow scope alone does not cover delivery constraints. MOE-04 requires a run record against agreed limits.

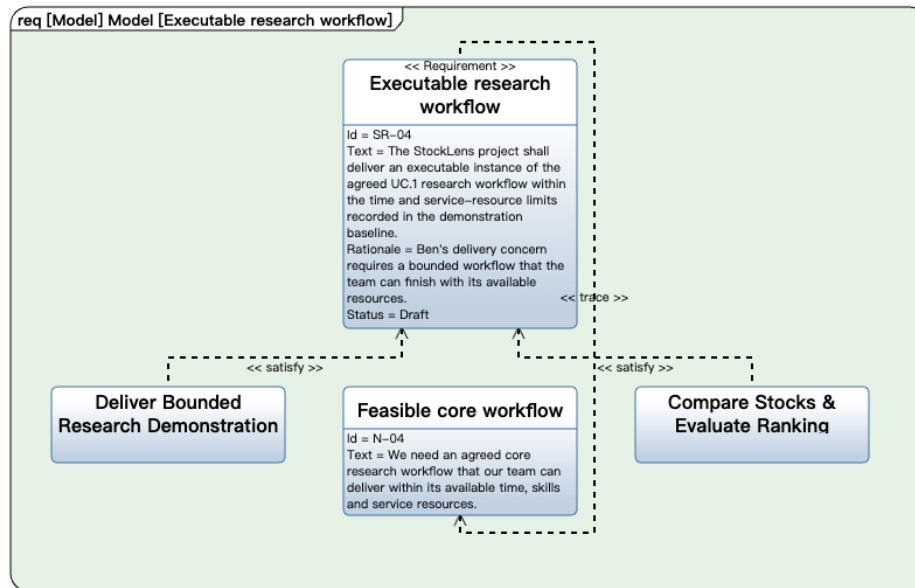


Figure 9: Native Innoslate Requirement Diagram for SR-04.

SR-05: Requirement traceability. SR-05 traces to N-05 and is allocated to A-05, Maintain Requirements Traceability. The project team maintains the record needed by evaluators such as Elena. MOE-05 inspects the correctness and completeness of the evidence chain; the existence of a relationship is insufficient on its own.

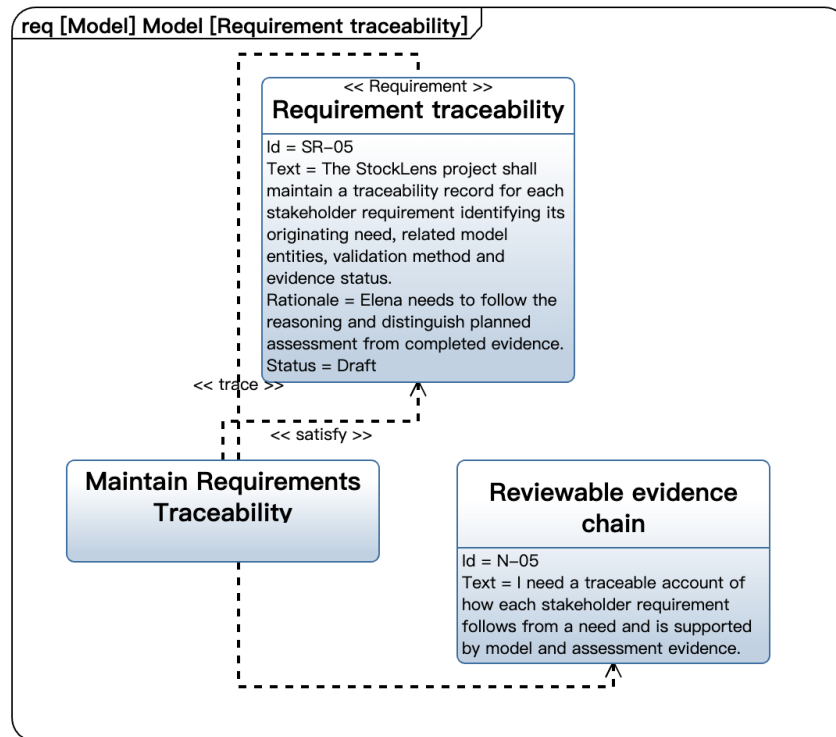


Figure 10: Native Innoslate Requirement Diagram for SR-05.

SR-06: Operation and transition guide. SR-06 traces to N-06 and is allocated to A-06, Maintain Operating and Transition Procedure. X.05 System Maintainer performs the action. Ben's continuity concern is assessed by another member executing the documented operation and transition procedure under MOE-06.

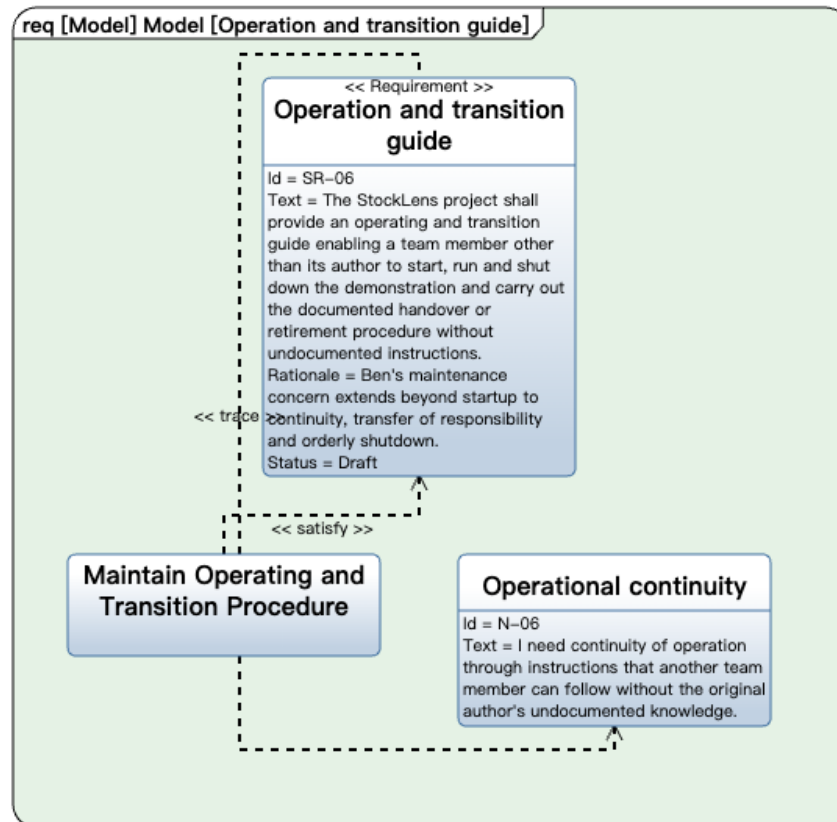


Figure 11: Native Innoslate Requirement Diagram for SR-06.

SR-07: Provider conditions. SR-07 traces to N-07 and is allocated to A-07, Control Service and Content Use. The project team and maintainer perform this work; external providers establish the applicable conditions. MOE-07 compares actual access, retention and display with the dated conditions, not merely the presence of a register.

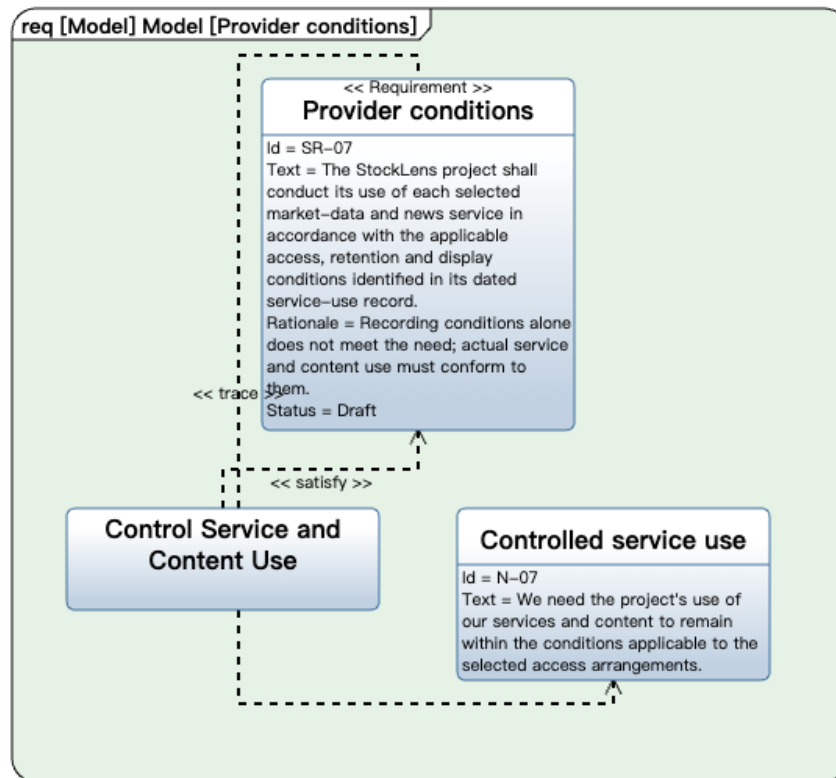


Figure 12: Native Innoslate Requirement Diagram for SR-07.

SR-08: Approved AI request schema. SR-08 traces to N-08 and is allocated to A-08, Control AI Analysis-Request Content, performed by StockLens System. UC.1.6 depends on this check before an external analysis request. MOE-08 still requires inspection of actual request payloads against the approved schema.

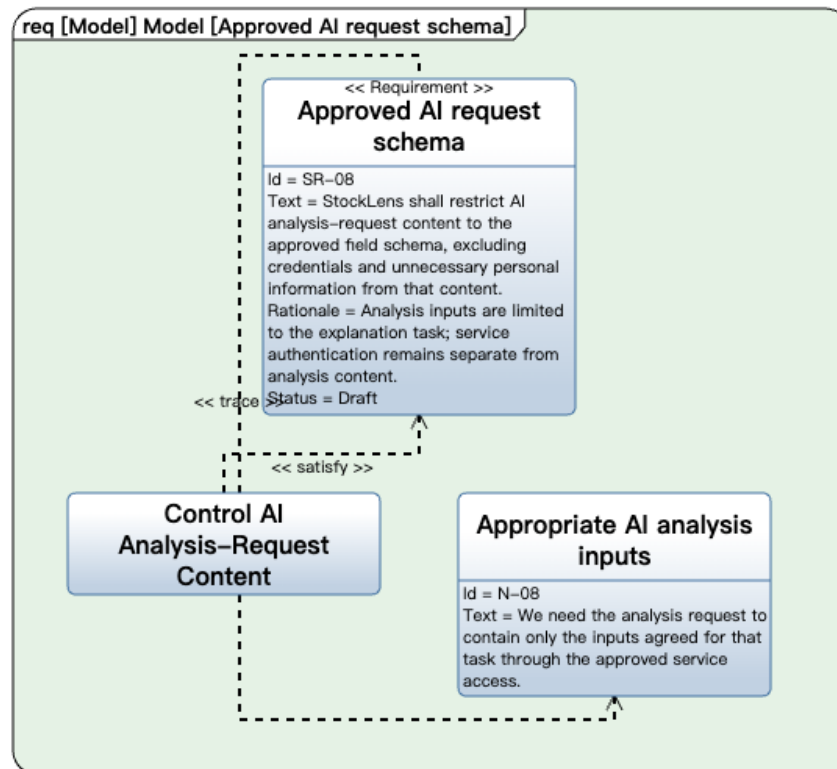


Figure 13: Native Innoslate Requirement Diagram for SR-08.

4.2 Requirements Traceability Diagram (Spider Diagram)

Figure 14 is the updated one-level Spider Diagram for SR-02. Its four direct links distinguish three relationship types: *traced from* N-02; *satisfied by* UC.1.6 and UC.1.7; and *sourced by* SL-SR. SL-SR is the maintenance document, not an implementing behavior. The same need and satisfying Actions appear in Figure 7.

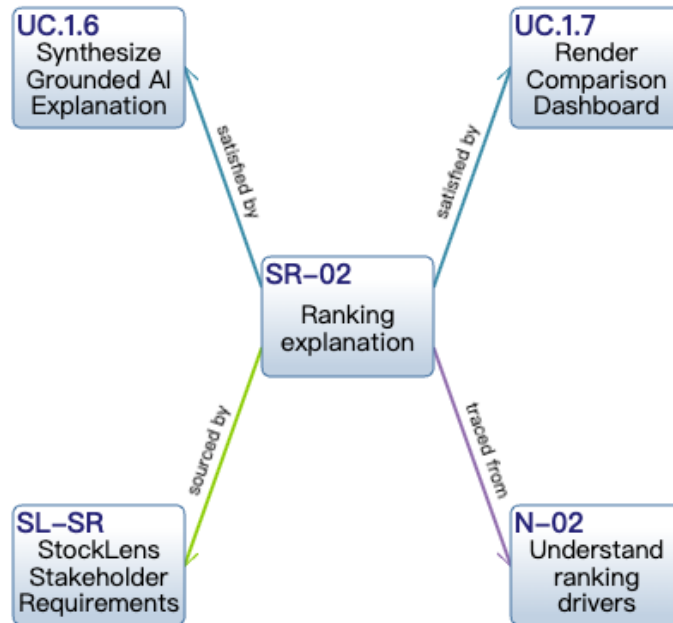


Figure 14: Native SR-02 Spider Diagram after the allocation correction.

The prose analysis connects this trace to Maya and Daniel’s concern, CTQ-02 and MOE-02; these additional items are not nodes in this one-level export. Table 14 provides the full-set relationship overview. Any future test-case relationship should distinguish a planned method from its execution result.

4.3 Requirements Traced to Related Entities (Matrix)

Table 14 records the model relationships created for this baseline. All eight requirements are sourced by SL-SR and trace back to their matching need in SL-N. The table distinguishes these stored links from the supporting evidence still to be produced. UC.1 is *Compare Stocks & Evaluate Ranking* (entity 185145); UC.1.6 (entity 186309) supplies the grounded explanation and UC.1.7 (entity 186503) renders the comparison. The Action names in the native figures correspond to these identifiers.

The stakeholder column preserves the source perspective from Table 9; it is not a claim that a stakeholder relationship has been added in Innoslate. ST-05 and ST-06 supply external constraints; the project team and maintainer control the project’s response. The revised allocations distinguish system provision from user inspection and distinguish delivery responsibility from the nominal workflow. All acceptance evidence remains pending.

Lifecycle and request-control allocations. The five draft Actions below describe responsible behavior and expected outputs. Their saved entity IDs are A-04: 219105, A-05: 216873, A-06: 217607, A-07: 217608 and A-08: 217609. Each is linked by the corresponding requirement’s *satisfied by*

relationship. Supporting records named in the descriptions are expected outputs, not a claim that those artifacts or validation results already exist.

Table 15: Draft Action allocations, responsibilities and assessment connections.

Req.	Action / responsibility	Behavior, records and assessment connection
SR-04	A-04: Deliver Bounded Research Demonstration Project team	Assemble and run UC.1 against the versioned scope, environment, time and service-resource limits. Retain the run record and deviations for MOE-04. Allocation does not claim a completed demonstration.
SR-05	A-05: Maintain Requirements Traceability Project team	Maintain one versioned record per SR with originating need, related entity IDs, validation method and evidence status. Reconcile changes across model and report; retain unresolved links explicitly. Output: traceability record assessed by MOE-05.
SR-06	A-06: Maintain Operating and Transition Procedure Maintainer	Record environment, dependencies, configuration and steps for startup, run, shutdown and the selected handover or retirement path. Include responsibility and treatment of retained records and service access. Output: guide; independent execution by another member supplies MOE-06 evidence.
SR-07	A-07: Control Service and Content Use Project team / maintainer	Review dated conditions for each selected service and control actual access, retention and display accordingly. Record use decisions and discrepancies; update controls when conditions change. Inputs: provider conditions. Output: service-use record and observations assessed by MOE-07. A conditions register alone does not satisfy SR-07.
SR-08	A-08: Control AI Analysis-Request Content StockLens behavior	Construct and check analysis content against the versioned approved field schema before sending it to the AI service. Exclude credentials and unnecessary personal information; keep authentication separate. Inputs: approved schema and candidate content. Output: permitted analysis payload; MOE-08 inspects actual requests.

The new Actions cover project-life-cycle work (A-04–A-07) and system request control (A-08). Their descriptions identify responsibility, behavior and expected outputs; they do not assert completed implementation. The names and IDs in this table match the saved relationships. Action timing defaults are not performance estimates or acceptance criteria.

Saved performers and stakeholder mapping. A-04 and A-05 are performed by ST-02 StockLens Project Team (entity 217610); A-06 by X.05 System Maintainer (199037); A-07 by both of these Assets; and A-08 by StockLens System (171150). Course Evaluators is represented by ST-03 (217611). The BMA Assets are reused: ST-01 corresponds to X.01 Research User, ST-04 to X.05 System Maintainer, ST-05 to X.02 Market Data Provider and X.03 News Provider, and ST-06 to X.04 AI Model Service. This report mapping avoids duplicating existing Assets merely to change their identifiers.

The evaluators' Asset represents an assessment role; it is not assigned implementation of the feature being assessed. A-04–A-08 are *Action* identifiers. They must not be confused with the BMA's separately scoped *assumption* identifiers, including assumption A-05.

Table 16 connects the same requirements to their stakeholder source, principal measure and planned evidence. It is a report-level association matrix, not an Innoslate matrix export. Items in its last column are assessment artifacts to retain; their listing does not assert that they already exist as model entities or completed results.

Table 16: Requirement associations with stakeholder perspectives and planned assessment evidence.

Req. ID	Need / source	Assessment	Related baseline or evidence artifact
SR-01	N-01 ST-01	MOE-01 User task; inspection	UC.1 comparison case; research universe; dated input set; indicator and scoring configuration.
SR-02	N-02 ST-01	MOE-02 User task; calculation review	Computed score breakdown; driver-selection rule; displayed explanation; task-answer record.
SR-03	N-03 ST-01	MOE-03 User task; source review	Displayed claims; supporting items and identifiers; publication, analysis and market-data dates.
SR-04	N-04 ST-02	MOE-04 Demonstration; review	UC.1 demonstration baseline; environment; confirmed time and service-resource limits; run record.
SR-05	N-05 ST-03	MOE-05 Inspection	Needs and Requirements documents; traceability register; model relationships; evidence-status records.
SR-06	N-06 ST-04	MOE-06 Demonstration; review	Operating and transition guide; configuration; selected handover or retirement path; independent execution record.
SR-07	N-07 ST-05	MOE-07 Actual-use inspection	Dated provider conditions; service-use record; observed requests, retained records and displayed content.
SR-08	N-08 ST-06	MOE-08 Payload inspection	Versioned field schema; service-access arrangement; recorded request set; redacted review evidence.

Completeness and change control. All eight requirements have a stored source-need link and a report-level assessment connection. All eight also have stored intended behavior allocations, including the five Actions in Table 15. This does not constitute a measured 100% result for MOE-05 or full satisfaction coverage. A changed need requires review of its requirement, CTQ, measure and method. Changes to scoring, provider conditions or the AI schema also require review of affected evidence. Update the model and report together, then regenerate the views and record the version, change reason and affected IDs using Table 13.

4.4 Intelligence Requirements Analysis

Innoslate Intelligence was rerun on 1 October 2026 after the revised allocations, performing Assets and AI-request dependency were saved. Its Requirement Analysis covered 73 requirements, including the pre-existing Airport Kiosk tutorial entities. It reported seven of nine applicable requirement heuristics passing at project level (77%). That aggregate is not a StockLens-only score, an individual Requirement Entity quality score, or a validation result.

The project-wide missing-satisfaction list decreased from 57 to 53 leaf requirements. Inspection of the expanded list confirmed that none of SR-01–SR-08 remained in it; the remaining entries belonged to the pre-existing tutorial requirement set. The four StockLens missing-link findings are therefore closed at the relationship level. The global ambiguous-word list did not include these eight SR statements. Neither result resolves the baseline definitions or demonstrates successful operation.

The rerun analyzed 37 Actions and 33 Assets. The project-wide count of Actions without a performing Asset decreased from 19 to 15; A-04–A-08 have saved performers. Other project and tutorial findings remain. UC.1.6 now has a saved dependency on A-08, but a dependency link is not executable control

flow or proof that requests are blocked correctly. The project-wide Action heuristic pass rate remains 50%; it is not a per-Action completion score.

Entity quality score. The saved requirements document continued to display a Quality Score of 0% for each of SR-01–SR-08 during the 1 October review. In the previously inspected SR-01 Entity view, the nine quality-review attributes (including Appropriate, Complete, Singular and Verifiable) remained set to No. The recorded 0% therefore must not be described as the Intelligence pass rate or proof that all statements failed semantic review. No positive quality-review attribute was asserted merely to improve the number.

Table 17: Observed entity scores, allocation findings and separate manual review items.

Req. ID	Entity score	Allocation issue	Manual review / disposition
SR-01	0%	Not flagged	Confirm eligible universe, scoring configuration and treatment of differently dated or missing observations.
SR-02	0%	Not flagged	Confirm the rule defining main computed drivers and use it consistently in explanation and assessment.
SR-03	0%	Not flagged	Confirm inspectable sources, claim support and the three date types.
SR-04	0%	Not flagged	Confirm UC.1 scope and time/resource limits before a repeatable acceptance decision.
SR-05	0%	Closed link	Inspect the traceability record and evidence status; allocation to A-05 is saved.
SR-06	0%	Closed link	Confirm transition path and responsibility; independently assess operation and transition together.
SR-07	0%	Closed link	Confirm dated conditions; compare actual service and content use with those conditions.
SR-08	0%	Closed link	Confirm the versioned field schema and inspect actual payloads against it.

The requirement text was preserved during this model entry and review. The team should resolve the remaining baseline definitions, record justified quality-review judgments, realize and verify the request-control gate, and rerun Intelligence after subsequent model changes. Automated checks supplement the proposed assessment plan; they do not establish user comprehension, operational continuity or provider compliance.

5 Summary

The report connects six stakeholder roles to eight needs and eight stakeholder requirements. The PIG prioritizes research users, the project team, evaluators and maintainers. The personas explain their comparison, understanding, review and continuity concerns; provider perspectives establish supporting constraints. Each requirement is linked to its originating need, a principal MOE and a proposed validation method.

The current Innoslate model preserves the BMA research workflow and records the revised requirement allocations. The native diagrams cover all eight need–requirement groups, and the SR-02 Spider view makes the source, origin and satisfying behavior explicit. The matrices record the same relationships and distinguish intended allocations from assessment artifacts. Performing Assets are assigned to A-04–A-08, and the AI explanation action depends on the request-content check. These changes improve consistency without treating relationship coverage as evidence of successful operation.

Validation remains planned. Baseline decisions, justified entity-quality review, executable request control and retained assessment results are still required before acceptance. The Product Build deliverables

in Lab Manual Section 3.5—SPEC.md, data and model contracts, need-named failing tests and an automatic build or CI check—are separate repository evidence; this report and its model diagrams do not establish their completion.

References

- [1] SYSEN 5151. *Foundations of Systems Engineering Lab Manual*. Version 3.0, September 7, 2026, Chapter 3, pp. 58–73.
- [2] Group 29. *Business or Mission Analysis: AI-Assisted Stock Research and Ranking Platform (Stock-Lens)*. September 19, 2026, stakeholder profiles, needs and proposed validation criteria.

Appendix A: Team Work Distribution

The six work packages below identify the lead responsibility for each part of this report. The allocation follows the sequence from stakeholder analysis to needs, requirements, assessment criteria and model traceability. Section leads are responsible for keeping their inputs and outputs consistent with the adjoining sections.

Member	Report sections	Lead responsibility and handoff
Yihan Zhou Part 1	2.1 Stakeholders; 2.3 Power-Interest Grid; 2.4 Prioritization	Establish the stakeholder register, explain power and interest placements, and identify priority roles. Provide consistent stakeholder identifiers and selection rationale to the persona and needs leads.
Shuxuan Wang Part 2	2.2 Personas	Develop the persona profiles, empathy maps and accompanying narratives for the prioritized roles. Connect their situations, concerns and goals to the stakeholder register and provide the scenario basis for needs refinement.
Henian Li Part 3	2.5 Stakeholder Needs; 2.5.1–2.5.3	Develop primitive needs, critical-to-quality considerations and effective needs. Preserve the stakeholder perspective and the PN-CTQ-N refinement links; provide the effective need baseline to the requirements lead.
Xinyuan Yan Part 4	3.1 Stakeholder Requirements	Translate effective needs into clear requirement statements, retain the originating need and rationale, and review statement consistency. Provide the requirement baseline and need-to-requirement mapping to the assessment and modeling leads.
Rosa Szurgot Part 5	3.2 Measures of Effectiveness; 3.3 Validation Criteria	Define a measurable MOE and a plausible validation method for each requirement. Maintain the integrated requirements/assessment tables and distinguish planned criteria from completed evidence. Coordinate identifiers and thresholds with Parts 4 and 6.
David Limmer Part 6	4.1–4.4 Requirements Management; report integration	Maintain the Innoslate needs and requirements documents, semantic relationships, Requirements and Spider diagrams, traceability matrix and Intelligence review. Integrate native exports with their explanations and coordinate the Introduction (Section 1) and Summary (Section 5) with the section leads.

Integration and review. Each handoff retains the agreed ST, N and SR identifiers. The section leads review their assigned content against the Lab Manual and the report’s upstream and downstream sections; the integration lead reconciles model exports, captions, cross-references and formatting. This table records responsibility allocation rather than measured hours or individual contribution percentages.